

GAPS.md — Gap & Experiment Registry (updated)

This registry is the living honesty instrument for the Find Us project. Every gap is tagged:

- [RES] — claim gap is a research/design finding (grounded in verified prior art audit).
- [SPEC] — gap is a specification/definition matter (resolvable by spec, not by measurement).
- [EXP:NOT-RUN] / [EXP:NOT-RUN] — gap closes only via designed experiments **that have NOT been executed** (no run artifact exists).
- [PROPOSED] — gap-closer is a proposed design (novelty candidate, unvalidated).
- [UNVERIFIED] — gap status depends on a source not independently verified.

Prior-art audit basis: `research/find-us-audit/01_prior_art_gap_audit.md`; current numeric/design ground truth: `RESEARCH_DUMP.md`. Last updated this research pass.

1. The Surviving Core Gap — Integration (G-INT)

Statement. Every atom in the proposed system exists as verified prior art in isolation: GPS-skeleton for cooperative positioning [RES]; cooperative estimation over BLE (SPAWN family, CoCoA line) [RES]; relative-coordinate meshes (CORELS/CLIPS family) [RES]; emergency BLE broadcast framing (OEPB, IgniRelay, CrisisConnect) [RES]; P2P emergency BLE patent coverage [RES]; degradation/uncertainty as continuous-variance factor-graph output [RES].

The gap: no single system in the verified pass integrates all of these under (a) a byte-capped legacy-BLE wire budget, (b) a mode-ladder of spatial degradation (metric→differential→topological→historical/ghost), (c) discrete confidence classes in place of continuous σ , (d) explicit mode-transition events, and (e) ghost-decay (nothing deleted, everything decays, ghosts never expire). Each integration component is [PROPOSED]; the *integration itself* is the contribution and is **unvalidated** [EXP:NOT-RUN]. [RES] (gap) · [PROPOSED] (closer) · [EXP:NOT-RUN] (validation).

Falsification: controlled experiments E1–E10 (below) that show any component performs worse or no better than single-phone GPS challenge the claim. Until E10 (2-phone real-device demo) passes, the system makes **no** "it works" claim. [RES]

2. Layer-by-Layer Gap Registry (12 sub-problems, priority-ordered)

Layer 1 — GPS Skeleton [RES] gap class

ID	Gap	Class	Closer
G-L1.1	No shared standard for <i>mode-level</i> GPS availability (silent skeleton↔denied transition; users see "10 m" when GPS just died)	[RES] gap · closer [PROPOSED]	L1 mode ladder: transition events (§3.2 ladder; E9)
G-L1.2	GPS-denied bootstrapping without a common frame (strangers aligning to a frame neither owns)	[RES]/[UNVERIFIED-de tail] gap · [EXP:NOT-RUN]	Frame-alignment protocol (E3/E5)

Layer 2 — Cooperative Estimation

ID	Gap	Class	Closer
G-L2.1	2-hop metric tier under resource cap (peer frame consistency without controversy)	[RES] → [EXP:NOT-RUN]	2-hop relative factor mesh (E1)
G-L2.2	Frame alignment without a common reference	[RES]/[PARTIAL]	Reciprocal-alignment (E5)
G-L2.3	Distributed vs centralized overhead under crowd scale	[SPEC]	Cost model (E4)

Layer 3 — Calibration [PROPOSED]

ID	Gap	Class	Closer
G-L3.1	Continuous self-calibration normally assumes a known frame; in emergency there is no anchor	[RES] gap	Mode-aware self-cal (E1/E2)
G-L3.2	Peer/reciprocal-validated calibration oracle — reciprocal inconsistency (disagreement vector between two peers) as the calibration oracle rather than a trusted anchor	[PROPOSED] · [EXP: NOT - RUN] · not found in prior art (empty prior-art result)	G-L3.2 falsification experiments (E6–E7) — this is the least-supported, most-original claim; explicitly unvalidated

Layer 4 — Uncertainty & Degradation

ID	Gap	Class	Closer
G-L4.1	Continuous σ is false precision in a mesh; discrete confidence classes + mode-ladder + explicit transition events not found as integrated boundary	[PROPOSED] · [EXP: NOT - RUN]	H5/E9; ghost-decay (H4)
G-L4.2	"Ghost" lifecycle — nothing deleted, everything decays; contradicting ghosts as honest alarm	[PROPOSED] · [EXP: NOT - RUN]	H4/E8

Layer 5 — Communication

ID	Gap	Class	Closer
G-L5.1	Byte-capped legacy-BLE wire ($\leq 23B$ iOS-safe) carrying a spatial model (not just a packet)	[RES] (bytes) · [EXP: NOT - RUN] (spatial-content)	E4 wire budget + DOC-06 frame
G-L5.2	2-hop emissions w/ error-tolerant FEC in byte cap; iOS background receipt (RPA, $\leq 23B$)	[RES]/[PARTIAL]	E4/E10

Layer 6 — Application

ID	Gap	Class	Closer
G-L6.1	Honest degradation UI — rendering "I don't know" with the same honesty as "I know"	[PROPOSED] · [EXP: NOT - RUN]	E9 (degradation-event rendering); H7
G-L6.2	Ghost UI + "head toward where I last saw them"; contradicting ghosts surfaced, not hidden	[PROPOSED] · [EXP: NOT - RUN]	E8

3. Honest Empty Results (novelty candidates with NO prior-art hit)

Candidate	Prior-art audit result	Load-bearing?
Peer/reciprocal-validated calibration oracle (G-L3.2)	No verified prior-art integration found	Yes — flagged [PROPOSED] · [EXP:NOT-RUN], never claimed as proven
Discrete confidence classes + mode-ladder + ghost-decay as integrated spatial-resolution boundary (G-L4.1)	No verified integrated hit	Yes — flagged [PROPOSED] · [EXP:NOT-RUN]
Byte-capped cooperative spatial exchange inside BLE budget (G-L5.1)	Components exist; integration absent	Yes — flagged [PROPOSED] · [EXP:NOT-RUN]

These empty results are the *survivable* integration gap; they are the honest, falsifiable contribution — and they are **design, not results**.

4. The Falsification Program (per RESEARCH_DUMP.md §8 / paper §6 — all [EXP:NOT-RUN])

ID	Falsifies / proves	Pass (example, [DESIGN-PARAMETER])
E1	RSSI calibration (log-distance)	σ within tier-1 target on $\geq 70\%$ devices
E2	Bearing (two-walk vs torso-scan consistency)	median bearing error $\leq$ target
E3	Compass drift across walks	within 2-hop tolerance
E4	Byte budget in 30-phone crowd (BLE)	trickle delivery $\geq$ target PRR
E5	Multihop error accumulation	error within tier bounds ≤ 2 hops
E6	Peer-validated calibration oracle (G-L3.2)	reciprocal inconsistency exposes drift
E7	Calibration honesty under adversarial drift	oracle says UNKNOWN, not wrong
E8	Ghost-decay navigation utility	ghost beats "unknown" in success rate
E9	Mode-transition event integrity	no silent mode flips; UI renders event
E10	2-device end-to-end demo (GATE)	searcher reaches victim using UI alone

GATE E10: until the 2-phone + relays real-device demo runs, the paper's "works" claims remain [PROPOSED], and every number is a [DESIGN-PARAMETER] estimate. [RES]

End of GAPS.md. A gap that is honestly empty beats a claim that is quietly full.